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Adaptive Stability-Plasticity Optimization for Factual Continual Learning in Large Language Models

 **ICML**

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Summary

This paper proposes MEWC-LLM, a continual learning framework aimed at reducing hallucinations in large language models by protecting factual knowledge during fine-tuning. The core idea is to adapt Elastic Weight Consolidation (EWC) to “factual truth preservation” using a Fisher-weighted penalty over a curated Factual Anchor Set

(FAS), combined with gradient conflict detection via cosine similarity and a Conflict-Aware Truth Preservation (CATP) gating mechanism. The authors claim improved factual stability and reduced hallucination without replay buffers, and outline a modular architecture (shared backbone with domain-specific adapters), but the experimental section presents mostly projected results rather than completed empirical validation.

Strengths

Technical novelty and innovation

- Reframes hallucination as a stability–plasticity (continual learning) failure rather than solely an inference-time retrieval issue, which is a compelling perspective.
- Integrates Fisher-based parameter importance (EWC-style) with gradient conflict detection (cosine similarity), leading to an adaptive consolidation coefficient; this hybridization is conceptually interesting.

Proposes a “truth-aware”

- Fisher computed over a curated FAS, aligning the consolidation objective with factual reliability rather than task identity.
- Introduces a structured CATP layer that modulates updates based on conflict severity, potentially offering a practical control knob for stability–plasticity trade-offs.

Experimental rigor and validation

- Clear definition of intended benchmarks and metrics (TruthfulQA, HaluEval, FEVER, HotpotQA, NQ, MMLU; HR, FCS, CFI, CAQ), laying out a reasonable evaluation plan.
- Provides an algorithmic loop for training with explicit steps for Fisher accumulation and conflict gating.

Clarity of presentation

- The optimization objective and conflict-detection mechanism are stated clearly with concise equations and pseudocode.
- The design of the FAS is described with sensible considerations (curation, staleness management, bias mitigation).

Significance of contributions

Addresses a highly important

- and timely problem—hallucination in LLMs under continual updates.
- If validated, a no-replay approach suitable for privacy-constrained deployments would be practically impactful.

Weaknesses

Technical limitations or concerns

- The proposed method largely reduces to EWC with a domain-specific anchor set plus a PCGrad-like conflict gate; the degree of novelty beyond combining known components is moderate.
- Reliance on a diagonal Fisher and top-K layer selection may miss important cross-parameter interactions in LLMs; the paper doesn't analyze the trade-offs or potential failure modes.
- Storing and operating with `g_prev` at scale (even in LoRA space) is non-trivial; memory and compute implications are not quantified.
- CATP's "reject update; RAG triggered" seems to contradict

the “without replay buffers” and “no external patches” narrative; the training-time/inference-time interplay is unclear.

Experimental gaps or methodological issues

- The paper mostly provides projected or illustrative results rather than completed, reproducible experiments. Claims such as “reduces factual drift by 52%” and “62.4% relative reduction in hallucination-inducing perplexity” lack end-to-end empirical backing with standard metrics, datasets, and baselines.
- Baselines and ablations are underspecified: comparisons to strong continual learning (SI, MAS, LwF, A-GEM, ER, DER++) and knowledge-editing methods (ROME, MEMIT, KE) are not executed.
- “Hallucination-inducing perplexity” is non-standard; its definition and relation to established hallucination metrics are unclear.
- The paper inconsistently refers to a GPT-2 backbone and later to LLaMA-3-8B/Mistral-7B; the final setup

and results are not consistently reported.

Clarity or presentation issues

- The expected-results table risks being read as actual results; this creates confusion and undermines credibility of the stated claims.
- The CATP thresholds are presented without principled derivation or tuning protocol; it is not clear how robust they are across domains.
- The flowchart and table formatting artifacts, while minor, occasionally distract from the core content.

Missing related work or comparisons

- Omits established continual learning regularizers and consolidation variants (Synaptic Intelligence, Memory Aware Synapses) and rehearsal-free hybrids; missing coverage of online EWC variants and KL-based stability constraints used in LLM tuning.
- Limited discussion of knowledge editing (ROME, MEMIT, FT-L, SERAC, KE) and how MEWC-LLM compares when the goal is preserving

and updating factual associations.

- Recent anti-hallucination approaches that use self-consistency, verifier models, or attribution-based methods are not contrasted empirically.



Detailed Comments

Technical soundness evaluation

- The objective in Eq. (1) maps directly onto EWC with the change that Fisher is computed on a FAS representing "truth." This is a reasonable reinterpretation but does not change the core math; the novelty is primarily in target selection and conflict-adaptive weighting.
- Cosine similarity-based conflict detection is a known device from multi-task learning (PCGrad, CAGrad). Applying it between gradients from new data and FAS is sensible, but more specifics are needed: is cosine computed in the full parameter vector or restricted to LoRA

parameters? Is it normalized per layer? How is stochasticity handled batch-to-batch?

- The CATP “reject” operation and RAG trigger require more formalization—e.g., what happens to the optimizer states (Adam moments) on rejection? How is stability ensured if frequent rejections occur?
- The FAS design is crucial. The paper does not quantify how large and diverse the FAS must be to protect broad factual knowledge, nor how to avoid overfitting to FAS distributions.

Experimental evaluation assessment

- The absence of completed experiments is the central limitation. A convincing evaluation would include:
 - End-to-end hallucination metrics on TruthfulQA, HaluEval, FEVER, HotpotQA, and NQ before and after domain adaptation, with CIs.
 - Catastrophic forgetting quantified on pre-adaptation tasks or held-out factual probes (e.g., PopQA/LongFact).

- Strong baselines: EWC, online EWC, SI, MAS, LwF, A-GEM, ER/DER++, PCGrad/CAGrad-style training, LoRA with KL regularization to a frozen teacher, and knowledge editing methods for targeted fact preservation.
- Ablations: with/without CATP; with/without Fisher; varying FAS size and domain composition; LoRA vs full fine-tuning; cosine threshold and α sensitivity sweeps; diagonal vs low-rank Fisher.
- Efficiency metrics: compute/memory overhead; wall-clock training time; impact on throughput.
- The notion of "hallucination-inducing perplexity" should be replaced or complemented with standard hallucination and factuality metrics (e.g., FACTS, TRUE, attribution faithfulness, entailment-based verifiers).

Comparison with related work (using the summaries provided)

- The paper rightly cites EWC and gradient conflict literature (PCGrad, CAGrad) and recognizes Fisher

approximations and modular architectures (LoRA, MoE). However, it does not empirically compare against these families nor against knowledge editing methods that also aim at fact preservation and targeted updates, which weakens the positioning.

- Recent surveys on RAG and editing are cited, but the claim that hallucination is primarily a learning-dynamics failure would benefit from evidence distinguishing model-internal alterations from retrieval/inference-time issues.

Discussion of broader impact and significance

- If validated, MEWC-LLM could materially improve reliability in privacy-constrained or edge deployments where replay is infeasible. It could help mitigate “domain bleed” in continual adaptation pipelines.
- Risks include encoding FAS biases as “truth,” ossifying models against legitimate updates or corrections, and potential brittleness when facts evolve (concept drift). Mechanisms for deconsolidation or time-aware

Fisher updates will be important.

- Ethical considerations around who defines “truth” and how disputes are handled deserve more discussion, especially for domains like law and politics.

Questions for Authors

1. Are the reported numerical improvements (e.g., 62.4% reduction in hallucination-inducing perplexity, 52% drift reduction) from completed experiments or illustrative projections? If they are real, please provide full experimental details, metrics, datasets, and variance across seeds.
2. How is `g_prev` stored and computed at scale? Is cosine similarity computed over all trainable parameters, per layer, or only within LoRA adapters? What are the memory and compute overheads?
3. How large is the FAS in your experiments, and how sensitive are results to FAS size, domain balance, and staleness? Please

provide ablations.

4. How do you handle update rejections in CATP with AdamW state (moments)? Do you reset optimizer states or freeze only a subset of parameters? What is the effect on convergence?
5. Can you compare MEWC-LLM to strong continual learning baselines (online EWC, SI, MAS, LwF, A-GEM, ER/DER++) and to knowledge editing methods (ROME, MEMIT, KE) on targeted fact preservation tasks?
6. How do you operationalize hallucination metrics beyond perplexity (e.g., TruthfulQA accuracy, FEVER evidence-supported entailment, HaluEval)? Please report standardized metrics with confidence intervals.
7. The paper mentions "RAG triggered" under high-conflict CATP. Is RAG used during training, inference, or both, and how does this align with the "no-replay, no external patches" claim?

Overall Assessment

The paper raises an important and timely question—how to retain factual reliability during continual adaptation—and offers a clear, well-motivated synthesis of EWC-style consolidation with gradient conflict gating. Conceptually, the framing of hallucination as a learning-dynamics failure is compelling, and the proposed mechanisms are plausible and practically oriented (e.g., LoRA-space gating, FAS curation). However, the submission's main shortcoming is the lack of completed, rigorous empirical validation. Many claims are presented as expected or projected, while some earlier statements imply realized gains, creating inconsistency. Without end-to-end results on standard hallucination/factuality benchmarks, strong baselines, and ablations, it is difficult to assess the true effectiveness and trade-offs of MEWC-LLM. I view the idea as promising and worthy of further development, but in its current form it falls short of ICML standards for experimental rigor. A future revision with comprehensive

experiments could be competitive.

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